

Build an Inert Module Characterization Bench

Review one declared fixture and preserve one exact result

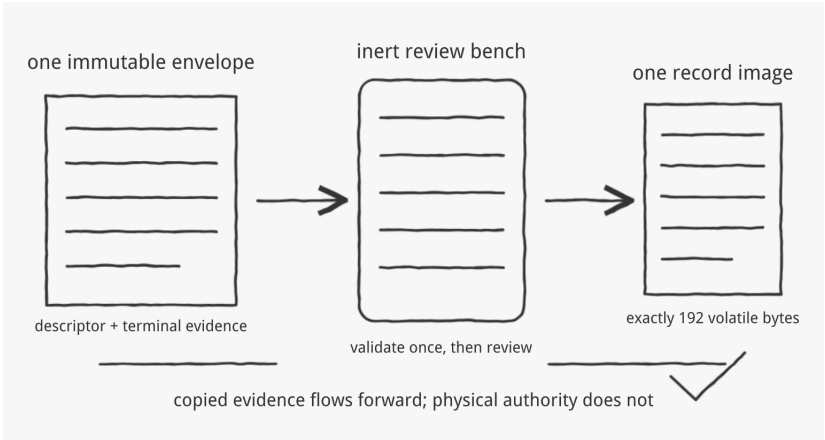
E0 COPIED-EVIDENCE REPLAY — NO POWERED MODULE OR ACCEPTANCE CLAIM

Lesson	072 — project	Time	180–240 minutes
Prerequisites	Lessons 070–071	Board	compile-only Mega replay
Input	one immutable characterization envelope	Observation	named result cells

Responsibility. `InertModuleCharacterizationBench` validates one atomic copied envelope, advances a bounded review script, and prepares one canonical 192-byte caller-owned volatile image. It does not identify, power, sample, calibrate, display, persist, accept, or declare safe a physical module.

1 Compose evidence without widening authority

Lesson 070 publishes a declared fixture contract. Lesson 071 characterizes copied samples for that exact descriptor and run. Lesson 072 consumes both together in one immutable `ModuleCharacterizationEnvelope`. It never reads two mutable child objects and hopes they describe the same instant.



Alternative text: a rough graphite envelope containing a descriptor and terminal evidence flows through an inert review bench into one record card labelled exactly 192 volatile bytes; a caption says copied evidence flows forward while physical authority does not.

The descriptor, evidence, and compact witnesses retain their complete authoritative fields. Domain-separated CRC-32 digests detect accidental collisions and support correlation, but a matching digest never replaces field comparison, identifies a marketplace board, or closes E1. Every 32-bit digest value is opaque: zero is valid and is never a sentinel, including in `expectedDescriptorDigest`.

E0 establishes	E0 leaves open
atomic descriptor/evidence correlation	exact board, carrier, and revision
bounded deterministic review	power, ADC, comparator, and potentiometer
canonical volatile record bytes	media receipt, durability, or certification
typed fault-dominant presentation intent	powered display or indicator behavior

2 Bind one exact declared fixture

`ModuleBenchConfig` fixes bench, envelope, record, descriptor, specimen, electrical-evidence, digest, and control-source revisions. The authoritative descriptor appears once inside the evidence. A changed descriptor, stale run, source mismatch, malformed enum, or digest mismatch rejects before session state or caller bytes change.

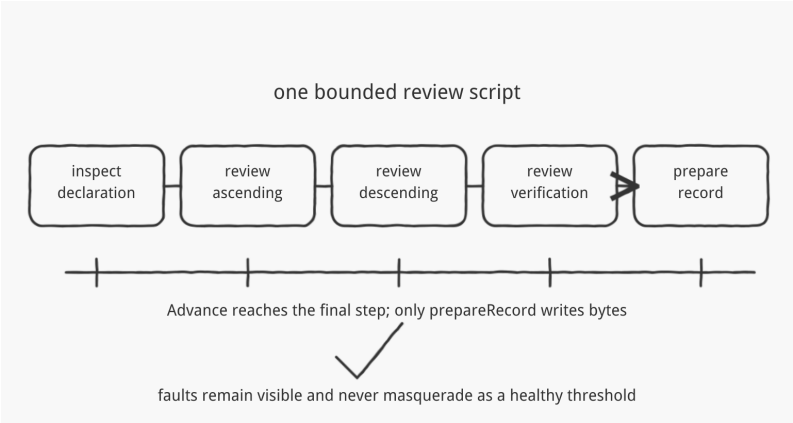
`beginSession()` applies lifecycle/configuration checks first, then structural envelope and descriptor/evidence validation, exact correlation, digest equality, and terminal admissibility. A structural, correlation, or digest failure rejects atomically even when another field resembles a producer fault. Only a fully correlated, digest-correct attributable rejection is admitted as fault-dominant evidence.

An `Unspecified` declaration may remain readable at E0, but it cannot be promoted into electrical admission. Light, sound, Hall, thermal, flame/radiant, Metal Touch, vibration, and obstacle-family labels are merely allowlisted copied-evidence shapes. They do not prove identity or connection. A flame/radiant label authorizes no flame, heater, ignition source, or hot target.

Suppose plausible evidence names a different electrical-evidence revision. Should the session begin? _____. Which information remains authoritative: the digest alone or the complete fields? _____.

3 Review exactly five steps

The bench starts `Ready`. One valid terminal envelope begins a `ScriptActive` session at `InspectDeclaration`. Each accepted `Advance` visits one next named step. There is no hidden acquisition loop and no duplicated begin/finish-leg protocol.



Alternative text: five rough graphite cards form one arrowed path: inspect declaration, review ascending, review descending, review verification, and prepare record. A note distinguishes advancing to the final step from the direct operation that writes caller bytes.

State	Advance	direct <code>prepareRecord()</code>
Ready	invalid	invalid
ScriptActive, before final	next named step	invalid
ScriptActive, final step	remains final	stage and commit image
RecordPrepared	invalid	idempotent, byte-identical
Fault or Shutdown	invalid	invalid

4 Keep chronology and control separate

A `ModuleBenchControl` binds source/configuration identity, session, sequence, observation time, command, and producer status. Evidence time, control time, sample sequence, run identity, lifecycle generation, and script step remain different quantities.

An unchanged duplicate is idempotent. A changed duplicate, regression, future control, stale control, required sequence gap, exact half-range ambiguity, or sequence exhaustion rejects without mutation. Producer failure dominates a plausible command. Direct `reset()` alone discards the saved compact snapshot, advances the nonzero generation, and returns to `Ready`. `Shutdown` is terminal and inert.

5 Preserve characterization meaning

Raw analog values remain integer codes in the declared ADC domain. Lesson 072 does not convert them into temperature, light, sound, gas concentration, distance, or touch capacitance. Comparator physical level remains separate from descriptor-interpreted assertion, so active-low is not inverted twice.

Ascending and descending brackets, guaranteed-active and guaranteed-inactive intervals, ambiguity, verification relation, chatter, no-transition, stale, disagreement, and endpoint-only rail results stay attributable to Lesson 071. Healthy rail points are ordinary evidence. `AtLowerRail` and `AtUpperRail` can arise only from endpoint-only finalization; neither means `OpenLike` or `ShortLike`.

Fault dominance. A plausible threshold cannot erase an identity, producer, age, rail, disagreement, or correlation fault. Diagnostic values may remain inspectable only when their invalid or diagnostic status remains explicit.

6 Prepare one normative volatile image

`ModuleCharacterizationRecordCodec` is the only encoder and decoder. The bench derives a compact semantic record, stages all 192 bytes, validates the result, and only then copies it into caller-owned output. No coordinator side path hand-encodes a second format.



Alternative text: a rough graphite record card is divided into ADCM framing, compact declarations, run evidence, and CRC-16 columns. Notes state that it is exactly 192 little-endian bytes with reserved bytes zero and is not a certificate or media receipt.

The image begins with ASCII **ADMC**, version, length 192, record and bench revisions, lifecycle/session identity, then compact descriptor declarations. It retains run identity and revision, three leg counts, ascending and descending brackets, three conservative intervals, comparator relation, witness and envelope digests, first/last sequences, terminal state/reason/status, script step, descriptor/envelope revisions, and control source/configuration. Reserved bytes are zero. CRC-16/CCITT-FALSE covers bytes 0–189 and occupies bytes 190–191.

The image intentionally omits terminal leg, current leg ID, observation times, control ordinals, directions, complete points, and full producer statuses. Those authoritative values remain in Lesson 071 evidence and are bound by its digest. The compact image cannot reconstruct the full envelope or a Lesson 071 result.

Decode checks exact span size, then framing, CRC, semantic/canonical fields, and only then assigns the staged output. Thus only a non-192-byte span is **BadLength**; wrong magic, version, or encoded length is **BadFraming**; a CRC mismatch is **BadIntegrity**; and a CRC-valid semantic defect is **BadSemanticValue**. Encode validates semantics before capacity, stages the image, zeroes reserved bytes, computes CRC last, and copies only on success. Compact-record digests are opaque and are not recomputed by the codec.

1. Change one reserved byte. Predict decoder outcome: _____.
2. Prepare twice after **RecordPrepared**. Byte difference: _____.
3. Reset after preparation. Is the former image durable evidence? _____.

Volatile boundary. The image records copied E0 review state during this process lifetime. It is not a hardware acceptance record, calibration certificate, conformance report, safety approval, filesystem commit, or power-loss durability claim.

7 Replay without touching hardware

The canonical Mega sketch constructs one descriptor and terminal evidence, forms their copied envelope, begins one session, advances the review script, prepares the record, and copies state, step, relation, fault, and **recordPrepared** into named volatile result cells.

1. **Predict:** write the expected bench state, script step, comparator relation, fault state, prepared flag, and first four image bytes.
2. **Observe:** compile for Mega 2560 and inspect the named volatile cells and caller-owned image in a debugger, simulator, or disassembly. Serial may support observation but is not required.
3. **Interpret:** compare the complete envelope, script progression, record framing, and fault precedence. Explain any disagreement before changing the fixture.

This is the E0 non-Serial observation path. It proves deterministic software publication separately from resource acquisition and released-line evidence. E0 owns zero pins, ADC channels, timers, interrupts, buses, supplies, displays, LEDs, power switches, or storage endpoints, so physical safe-state evidence remains open.

8 Publisher verification and open acceptance

Host evidence	Strict lifecycle, correlation, chronology, fault collision, codec golden-vector, corruption, canary, reset, and shutdown tests pass.
Mega evidence	Synthetic replay compiles; exact linked no-LTO flash/static SRAM, stack, coordinator, 192-byte image, simultaneous composition, and residual SRAM evidence are measured.
Document evidence	PDF metadata, extraction, page count, monochrome, visual classification, and rendered-pencil review pass.
Physical evidence	Open: exact specimen, supply, ADC/comparator endpoints, pulls, adjustment control, observation endpoint, and measured bench result.

The canonical repository Mega build measures 24,860 B flash and 1,998 B static SRAM ordinarily. The isolated no-LTO build measures 27,354 B flash plus 2,002 B static SRAM. Measured stack is 740 B, the bench object is 436 B, each image is exactly 192 B, staged plus destination images are exactly 384 B, and residual SRAM is 5,322 B. The no-LTO flash result exceeds the 24 KiB target but remains below the independently reviewed 32 KiB hard limit. All other reported measures pass their targets and hard limits. This reviewed software target miss is not hidden, and passing the hard gates cannot be presented as physical acceptance.

9 Draw the powered acceptance boundary

There is no wiring diagram or formal schematic at E0. Before any powered claim, one exact specimen needs traceable markings and revision, primary electrical evidence, supply and signal limits, output-stage and pull requirements, ADC reference/domain, warm-up and settling declarations, potentiometer direction, and an electrically authoritative formal schematic.

The E1 bench card must separately record prediction; where and when to observe a non-Serial result; interpretation; resource-acquisition evidence; safe-state evidence; ascending, descending, and verification runs; comparator/analog agreement; rail and chatter outcomes; and acceptance disposition. A compiled sketch, host fixture, family name, or plausible curve is not that record.

No heater, flame, laser, ignition source, pyrotechnic control, launcher, unknown-protocol replay, energized moving load, or unattended hazard is authorized. An unidentified board stays unpowered.

10 Exercises and acceptance reasoning

1. Pair one descriptor with terminal evidence from another run. Prove every state value and caller byte remains unchanged.
2. Exercise both comparator polarities and both ramp directions. Show where physical level and interpreted assertion remain distinct.
3. Drive each named script step, then prove **Advance** alone never writes the record image.
4. Corrupt framing, enum, reserved, payload, and CRC fields. Confirm the decoder never partly publishes a record.
5. Replay chatter, no transition, a healthy rail point, endpoint-only rail finalization, stale evidence, and disagreement. Explain why none becomes a physical root-cause diagnosis.
6. List the exact E1 evidence needed before connecting one declared module, and distinguish resource acquisition from safe-state proof.

Project result. You have composed one declared descriptor and one terminal characterization into a bounded, fault-dominant inert review and exact volatile image without smuggling hardware identity, calibration, acceptance, or persistence into E0.

Stop at E0: every powered specimen, wiring, formal schematic, presentation endpoint, persistence mechanism, and physical acceptance result remains open work.